



Learning disciplinary literacy teaching: An examination of preservice teachers' literacy teaching in secondary subject area classrooms[☆]



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HIGHLIGHTS

- We describe early attempts of preservice teachers' disciplinary literacy teaching.
- All novices employed literacy teaching moves in their first teaching attempts.
- Some novices aligned disciplinary inquiry with literacy support.
- We offer implications for the study and design of teacher education.

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ABSTRACT

In this study, we sought to articulate observable features of literacy teaching in subject area classrooms and to explore possibilities for teacher education design focused on disciplinary literacy teaching. We analyzed video records of practice of two cohorts of secondary undergraduate teaching interns ($N = 60$) who were enrolled in their first semester of a three-semester university-based program. To conduct our analysis, we developed a coding scheme that reflected sociocultural theories framing inquiry and literacy as social practice. We found that interns, with support, attempted important moves related to disciplinary literacy teaching.

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Supporting adolescent literacy learning is important because as young people move through school the texts become more complex, the concepts become more abstract, and the purposes for reading and writing become more specialized (Snow & Moje, 2010). Comprehension instruction alone cannot meet these demands (Lee & Spratley, 2010). Accordingly, student learning standards in many educational systems worldwide now foreground advanced literacy learning in the subject areas, such as learning to evaluate claims and learning to use language purposefully (e.g., *Australian Curriculum*; the United States's *Common Core State Standards*).

Central to adolescent literacy teaching and learning is *disciplinary literacy*, or the social and problem-based work with texts that enables the critique and production of knowledge. Disciplinary literacy teaching approaches focus on surfacing the specialized and

distinct literacy practices of each academic discipline (e.g., chemistry) and helping students learn to use them. More than teaching children to be “mini scientists” or “mini historians,” many proponents argue that disciplinary literacy teaching and learning has an important role to play in promoting equity and justice (e.g., Lee, 2007). This is because specialized and critical literacy skills may enable people to better negotiate the literacy demands of their lives.

Although there are a variety of compelling reasons for attending to adolescents' disciplinary literacy learning, there are pressing questions about how to best support preservice teachers' disciplinary literacy teaching (e.g., Carlson, 2015; Conley, 2012). In this study, we analyze and describe initial attempts by novice secondary preservice teachers (i.e., “interns”) to provide literacy instruction within a restructured university-based secondary teacher education program that emphasizes literacy teaching in the academic disciplines. Our research question was: Within one secondary teacher education program, what was the nature of preservice teachers' earliest attempts to enact disciplinary literacy instruction in subject area classrooms?

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1. Literature review

1.1. Disciplinary practice

We work from a sociocultural perspective, wherein we view all literate practice as occurring within nested social and cultural contexts (Lankshear & Knobel, 2007; Street, 1984). Reading, writing, and reasoning are never detached from the purposes, communities, norms, and practices of the discourse communities in which they occur (Gee, 1990).

Academic disciplines (e.g., biology, history) are special types of social and cultural contexts (O'Brien, Stewart, & Moje, 1995). Members of academic disciplines share knowledge, practices, and values that enable them to communicate and construct knowledge over space and time with one another. Specific differences in practices, including practices with texts, mark the boundaries of disciplines (Goldman et al., 2016). Historians, for instance, regularly seek out and consider primary sources in specialized ways. To make sense of texts when engaged in historical analysis, historians employ disciplinary literacy practices such as reading primary sources, determining significance, corroborating, contextualizing, and sourcing (Bain, 2006; Leinhardt & Young, 1996; Wineburg, 1991a; 1991b; 1998). Natural scientists (e.g., chemists) employ different literacy practices as a regular part of their work. They consider multiple representations of data patterns using texts such as lab notes, and they also develop and use models and translate data across multiple forms and symbol systems (Goldman & Bisanz, 2002; Shanahan, Shanahan, & Misichia, 2011). Mathematicians engage in detailed reading and rereading of focal texts to identify errors and inconsistencies; this is because precision is so important for all aspects of mathematical representation (Bass, 2006; Shanahan et al., 2011). Still, mathematicians tend to understand that mathematical knowledge is socially constructed within a community of practice (Draper & Siebert, 2004; Siegel & Borasi, 1994). Literary scholars, too, work with texts in specialized ways (Lee, Goldman, Levine, & Magliano, 2016). Literary scholars seek patterns within and across texts, consider histories of use and other contexts, generate literary puzzles to pursue, and recursively construct meaning, guided by their goals to develop literary interpretations (Rainey, 2017; Reynolds & Rush, 2017).

1.2. Disciplinary literacy teaching

Moje (2007) review of the extant literature on subject-matter teaching and literacy teaching found four distinct conceptions of disciplinary literacy teaching: cognitive, epistemological, linguistic, and cultural. Work focused on *cognitive literacy processes* has sought to understand the various cognitive strategies and resources involved in reading and writing subject area texts and to develop approaches for supporting students' comprehension and subject-matter learning. Such approaches include the use of graphic organizers (e.g., Ogle, 1986) and supportive routines for reading and classroom talk (e.g., Palincsar & Brown, 1984). Work focused on *epistemological processes* has examined how members of an array of disciplines think, including how their purposes shape their processes for learning from and communicating with texts (e.g., Hand, Hohenshell, & Prain, 2004; Wineburg, 1991a, 1991b). Work on *linguistic processes* has examined how language and subject-matter learning are interrelated. Notably, scholars working from a systemic functional linguistics perspective (Halliday & Matthiessen, 2004) have analyzed and sought to teach students features of privileged academic discourses, including features of genre, grammar, technical language, and other linguistic choices (e.g., Schleppegrell, 2004). Work focused on *cultural practices* has sought to offer students support as they navigate "across the different

cultural, discursive, or linguistic communities of secondary schooling and of their everyday lives" (Moje, 2007, p. 27). This culturally focused line of work (e.g., Moje et al., 2004; Gutiérrez & Rogoff, 2003) seeks to value students' everyday knowledge while teaching them to consider privileged disciplinary discourses and practices as human constructions developed by people over time, as opposed to being static or immutable.

Moje (2007) concluded the review by calling for increased integration of these four broad approaches to disciplinary literacy, arguing that teaching for social justice requires it. When integrated, these four approaches would move towards equipping youth to critique and construct knowledge and to challenge the oppressive structures of society and the disciplines.

As a first step in representing disciplinary literacy teaching, Moje (2015) advanced a four-part heuristic that would integrate the prominent lines of relevant scholarship in practice. The heuristic, based in her review, centers on *engaging* students in work that aligns with the problem- and text-based work of disciplinarians (rooted most specifically in the epistemological lines of scholarship); *eliciting* students' understandings and *engineering* students' learning opportunities so that they are able to successfully accomplish classroom tasks and learn disciplinary practice (rooted in the cognitive lines of scholarship); and supporting students to *examine* and *evaluate* words and other forms of representation and ways with words and representations within and across discourse communities (rooted in the linguistic and cultural lines of scholarship, respectively) (Moje, 2007).

Based on Moje's (2007; 2015) work, we understand *disciplinary literacy teaching* as *teaching for social justice*. It is activity which

- *engages* students in disciplinary practices and literacy practices of a focal discourse community, especially ways of using texts within cycles of inquiry,
- *engineers* or scaffolds students' access to those practices, norms, and values, including eliciting what students know and bring to their work,
- supports students as they *examine* aspects and functions of language, linguistic choices, and genre features, and
- supports students as they *evaluate* "ways with words" and other forms of representation, including critically considering others' use of language and literacy and making decisions about their own literacy practice across domains.

This framework enables us to theorize what disciplinary literacy teaching may include at the lesson level. Fundamentally, it would include a clear disciplinary problem or purpose and support students' reading/writing in the service of that purpose, but it would also move beyond uncritical apprenticeship toward supporting students' critique of and navigation among the many discourse communities of their lives.

1.3. Empirical research on disciplinary literacy and preservice teacher education

Since Moje (2007) review, there have been important advances in the integration of cognitive, epistemological, linguistic, and cultural dimensions of disciplinary literacy teaching, particularly regarding professional development and curricular support for in-service teachers (c.f., De La Paz et al., 2014; Duhaylongsod, Snow, Selman, & Donovan, 2015; Greenleaf et al., 2011; Lawrence, Crosson, Paré-Blagoev, & Snow, 2015; Lee & Goldman, 2015; Litman et al., 2017; Reisman, 2012). Together, these and other studies have contributed mightily to the field by developing and implementing innovative curricular approaches in secondary classrooms and demonstrating the impact that disciplinary literacy

teaching can have on student learning.

Despite these and other advances, questions remain regarding how preservice teachers and those new to disciplinary literacy teaching may be supported to learn such approaches. How may teacher education institutions prepare novice teachers for disciplinary literacy teaching, the kind of teaching that even many teacher educators are unlikely to have experienced as students themselves? What are typical patterns in preservice teachers' initial attempts to design and enact disciplinary literacy teaching?

To date, research into preservice literacy teaching and teacher education has largely centered on documenting and shaping novices' beliefs or knowledge about teaching literacy, with data typically collected from a methods course and analyzed by the methods instructor (e.g., Colwell, 2016; Colwell & Anderson, 2016; Hsieh, 2017; Masuda, 2014; Olson & Truxaw, 2009; Park, 2013; Pytash, 2012; Siffrinn & Lew, 2018). A review of 50 years of teacher education and literacy research underscored the overwhelming emphasis on preservice teachers' beliefs, knowledge, and dispositions related to literacy teaching, noting some indications that preservice teachers' beliefs do not always transfer into their classroom implementation, and also noting a general theme of subject area preservice teachers' resistance to literacy teaching (Scott, McTigue, Miller, & Washburn, 2018). Scott et al. (2018) concluded with a call for more research, including research that would extend what is known about novice teachers' application of literacy teaching in subject area classrooms.

Some recent work has sought to study the subject-specific teaching of preservice teachers within teacher education contexts designed to support their learning of teaching practice (e.g., Kavanagh & Rainey, 2017; Kloser, Wilsey, Madkins, & Windschitl, 2019). In one study, Reisman et al. (2019) investigated teacher candidates' facilitation of historical text-based discussions. The design involved 32 primary and secondary history preservice teachers prepared at one of two different universities. The authors conceptualized historical text-based discussions as including four component parts: a) engaging students as sensemakers; b) orienting students to the text; c) orienting students to the discipline; and d) orienting students to one another (Reisman et al., 2018). Results showed that although the preservice teachers consistently employed many of the discrete moves of facilitating discussion, their facilitation often remained focused on students' comprehension rather than on questions of history within a disciplinary frame. In the second year of data collection, preservice teachers were introduced to the conceptual framework for historical text-based discussions; that year, participants tended to more frequently integrate these component parts in their enactments. In a follow-up study, the same authors examined how teacher educators worked with preservice teachers to "attend to the structure of the discipline" and to "construct and verify knowledge in disciplinarily appropriate ways" (Kavanagh et al., 2019, p. 1).

Together, these recent studies suggest the complexity of learning to enact disciplinary literacy teaching for novices and the promise of supporting them to do so. Our work builds on this scholarship by exploring how preservice teachers learn to design and enact disciplinary literacy teaching in a teacher education program organized to advance approaches to problem-based, text-based, and student-centered subject-area instruction.

2. Method

Guided by our sociocultural perspectives and Moje's (2007; 2015) arguments for social justice teaching as integrating four distinct lines of literacy and subject-area scholarship in practice, we asked: Within one teacher education program, what was the nature of preservice teachers' earliest attempts to enact disciplinary

literacy instruction in secondary subject area classrooms? Heeding Scott et al.'s (2018) call for studying preservice teachers' literacy teaching in subject area classrooms, we focused our analysis on novices' classroom-based enactments.

2.1. Participants

Our study included two cohorts of undergraduate¹ teaching interns enrolled in their first semester of a three-semester university-based secondary teacher education program that was redesigned to prepare novices to provide disciplinary literacy teaching. Cohort I was composed of all 37 interns who enrolled in the program in one semester. Cohort II included 23 of the 32 interns who enrolled in the program in the following year. Nine interns were excluded from our analysis because they did not collect viewable teaching videos.² Most interns entered the teacher education program as third-year undergraduates having partially completed their coursework in their respective majors (e.g., physics). See Table 1 for additional information about participants.

2.2. Program context

At the time of data collection interns were enrolled in an undergraduate secondary teacher education program at a university in the Midwest United States. Since 2000, program leaders have redesigned and studied many aspects of the program to facilitate interns' professional learning (Bain & Moje, 2012). Many of the redesign efforts were focused on advancing disciplinary literacy teaching. One redesign effort involved the design and goals of the program's content literacy course, a state requirement for preservice teachers seeking initial teaching certification. Before redesign, the course exposed interns to principles of comprehension instruction and to generic strategies for supporting students' learning from texts. Because the course included interns from a range of subject areas, it was difficult to support interns' learning of deep subject-specific literacy teaching and learning. After redesign, interns began to take a subject-specific literacy course. This reform meant that all science interns took one section of the foundations of literacy course designed to enable their investigation of ways of integrating science and literacy teaching, with all mathematics interns taking a separate section of the course, and so on. Course goals involve supporting interns as they develop a vision for disciplinary literacy teaching within their own academic domain that is socially and culturally sensitive and student centered.

A second reform effort involved intentionally partnering with veteran mentor teachers who shared program leaders' vision of disciplinary literacy teaching and tying an 8-hour per week field experience to the first semester teacher education coursework. Instead of a one-to-one placement model in which one intern would be placed with one mentor teacher and then teach their first lesson independently, the program began to place up to four interns with a single mentor teacher. This shift enabled interns to complete most of their course assignments together as a small group in the context of their mentor's classroom. In addition to extending opportunities for peer learning, small group field placements enabled more programmatic control over the quality of the teaching placements and the alignment of interns' practical experiences with program goals.

¹ Participants do not include interns seeking foreign language or elementary certification or those enrolled in the Master's with certification program.

² Video records were not systematically missing. Two math groups and one English group did not collect viewable videos due to issues with technology or video quality.

Table 1
Undergraduate secondary teaching interns by cohort and content area (N = 60).

Teaching Certification Area	Typically Associated Undergraduate Majors	Cohort I	Cohort II
English language arts (ELA)	English, American literature, composition	15	8
Mathematics	Mathematics, computer engineering	7	5
Social Sciences	History, political science, economics, psychology	11	6
Natural Sciences	Biology, chemistry, earth and environmental sciences	4	4
Total		37 (Female = 25; Male = 12)	23 (Female = 15; Male = 8)

A major project in the foundations of literacy course was the planning, enactment, and reflection on a single literacy lesson within their mentor teachers' classrooms. The expectations of the assignment were for interns to design a text-based lesson plan to support students' literacy learning situated within their mentor teacher's larger inquiry-driven unit. When designing their lesson, interns also formally considered: (a) the cultural, social, and literacy backgrounds of students in their placement classroom; (b) the literacy needs of students based on a suite of assessments; (c) the concepts and disciplinary knowledge undergirding the focus of the lesson; and (d) the potential literacy challenges posed by their lesson's focal text(s). Course instructors supported each aspect of interns' lesson planning in their foundations of literacy course; instructors also guided interns in rehearsals of prioritized moments of their lesson plans in advance of their actual teaching. Mentor teachers and field instructors played a range of roles during enactment to ensure students received high quality instruction (e.g., circulating to small groups, intervening in particular moments). After the lesson, interns reviewed their videorecording and student work to reflect on and articulate professional next steps.

2.3. Data collection

Our teacher education program routinely collected videos of interns teaching in their mentor teachers' classrooms. Videos were stored in a private (password-protected and encrypted), university-owned virtual channel for use by program faculty and interns. For this study, we accessed every video representing the first teaching attempts of interns in two cohorts. Video data are appropriate for our inquiry because we are focused on the nature of interns' initial enactment, and videoed lessons reveal the instructional practice of preservice teachers in classrooms with students. Our data included 23 videoed enactments collected from 19 groups of interns (see Table 2). Groups co-planned and typically co-taught the lessons with extensive feedback and support from their instructors and mentor teachers. In Cohort II, math interns co-planned but individually taught lessons in their mentor teacher's classroom, so the total number of minutes coded is higher for the math groups in that year. Variations are due to school and classroom differences among mentor teachers (e.g., differences in length of class period) and slight differences in how literacy methods instructors approached the assignment.

2.4. Data analysis

We employed provisional coding in our data analysis (Miles, Huberman, & Saldaña, 2020; Saldaña, 2016). Provisional coding is a technique that begins with an initial list of researcher-generated codes based on previous scholarship. Provisional coding is both deductive and inductive, enabling researchers to revise, delete, or expand the starting list of codes. Our use of provisional coding enabled us to explicitly build upon existing conceptualizations of disciplinary literacy teaching while also ensuring that we would, in

the end, describe the instruction reflected in our data set. Although we ended up organizing our data according to Moje's (2015) four-part heuristic, our process was emergent and did not begin with that decision. Instead, we began with a short list of anticipated literacy teaching moves that we might identify in the video data based on the literature and on our knowledge of the teacher education program. These included codes such as "supporting vocabulary use" and "activating prior knowledge." As we approached the data, we made careful notes about what we observed, including the extent of inquiry-driven and text-based instruction occurring in each video and variation among each of the four academic domains and among middle and high school contexts. We applied our drafted codes and refined our list by asking ourselves what they were not yet capturing. We generated codes such as "administering assessment" and "leading discussion" to indicate the activity structures evident; "analyzing bias and perspective in language" to name the close reading work that we observed; "using a graphic organizer" and "modeling a strategy" to name the literacy support we observed; and "naming norms of the disciplinary community" to represent the social and cultural apprenticeship we observed.

Next, we sought to generate a thematic scheme to describe the literacy teaching we observed. We tested various ways of clustering our codes and subcodes, including an initial scheme of *framing* (i.e., creating an inquiry frame by posing a question or problematizing a concept); *naming* (i.e., explicitly naming literacy strategies and practices, and disciplinary norms and practices); and *doing* (i.e., reading, writing, and constructing knowledge). Subsumed under each were a set of codes that described types of framing, naming, and doing. This initial scheme seemed to capture some of the major patterns we observed, and it enabled us to consider similarities and differences across the various lessons (e.g., how did the lessons that only included *naming* differ from those that included all three dimensions?). Yet, this initial scheme did not foreground the language- and text-based aspects of the work quite as clearly as we needed to answer our research question, and it was not yet fully integrated with extant scholarship.

We revisited Moje's (2007; 2015) syntheses of disciplinary literacy scholarship and began to sort our codes along the 4Es heuristic (Moje, 2015). We found that in some cases Moje's work gave us more precise language for what we were noticing and seeking to label. For example, "engaging" students in disciplinary activity includes framing a problem, working with data, and communicating claims, along with other activities. These became the foundation for our labeling of this dimension of instruction.

Yet, there were still ways in which our ultimate coding scheme would differ from the specific components identified by Moje (2015) because we sought to understand and name the observable aspects of disciplinary literacy teaching present in our data. In analyzing the videos, we noted that a typical way that interns established an inquiry frame was to pose a specific inquiry question. The code "posing a persuasive question" became a key look-for as we determined whether a lesson included the engaging

Table 2
Video data collected.

Intern Group	Grade Level Taught	Content Area	School Type	# Interns/Group	# Episodes Coded	# Minutes Coded
Cohort I						
A	8	ELA	Public suburban	4	1	95
B	8	ELA	Public suburban	4	1	80
C	11	ELA	Public rural	4	2	85
D	11	ELA	Public rural	3	1	100
E	8	Social studies	Public charter urban	4	1	80
F	8	Social studies	Public charter urban	4	1	82
G	8	Social studies	Public charter urban	3	1	44
H	10	Natural sciences	Public suburban	2	1	75
I	10	Natural sciences	Public suburban	2	1	75
J	12	Mathematics	Public exurban	3	1	53
K	12	Mathematics	Public exurban	4	1	58
Cohort II						
L	9–11	ELA	Public suburban	4	1	34
M	9–11	ELA	Public suburban	4	1	48
N	9	Social studies	Public suburban	3	1	82
O	9	Social studies	Public suburban	3	1	73
P	6	Natural sciences	Public suburban	2	1	59
Q	6	Natural sciences	Public suburban	2	1	56
R	9	Mathematics	Public suburban	3	3	169
S	9	Mathematics	Public suburban	2	2	137
Total Episodes and Time of Coded Teaching Practice					23	1485 min (24.75 h)

dimension. Although posing an inquiry question is related to framing a problem, we separated these into two different codes and defined framing a problem as activity such as problematizing a concept. Moving back and forth among the scholarship and our data analysis in this way ultimately enabled us to align our coding with the extant literature and theory while also ensuring that the scheme described our data well.

Throughout our process, we acknowledged that our coding scheme would not represent all aspects of disciplinary literacy teaching. We do not include teacher reasoning, aspects of planning or debriefing/reflection, or dispositional aspects of teaching. The choice is not borne out of a disregard for these important aspects of teaching but rather a decision to focus as closely as possible on observable moves of disciplinary literacy teaching in enactment. Further, because our participants were preservice teachers, we expected that our coding scheme would only partially describe the complexity of many veterans' disciplinary literacy teaching.

We applied our final coding scheme to both waves of intern teaching video collected in our program. See Table 3 for full coding scheme. All videos were coded by two members of our research team ($k = .89$). Additionally, a subset of coded videos were independently reviewed by a visiting scholar who was a doctoral candidate pursuing her own research on disciplinary literacy teaching in secondary contexts. The visiting scholar's coding was highly consistent with our team's coding. We reconciled all instances of disagreement through discussion.

3. Findings

We sought to understand the nature and extent of initial attempts of disciplinary literacy teaching in one secondary teacher education program. Based on our analysis of our data, we assert that 1) interns consistently exhibited literacy teaching moves in their videoed enactments; and 2) holistically, their attempts revealed a range of low to higher integration of the four dimensions of disciplinary literacy teaching.

In what follows, we report overall patterns we detected in the videoed lessons using Moje's (2015) four Es: engaging, engineering, examining, and evaluating. Then, we explore patterns in combinations of the Es at the lesson level.

3.1. Enacting the component parts of disciplinary literacy

Interns' teaching videos represented a range of the four dimensions of disciplinary literacy teaching. Engineering was evident in all 23 videos and the most frequent component. Second most frequent was engaging (12 videos), followed by examining (9 videos), and evaluating (3 videos). Table 4 shows the videos organized by these patterns.

3.1.1. Extent of engaging with the practices of the discipline

Interns in 12 videos (of 23) offered students opportunities to engage in disciplinary inquiry. In these videos, interns offered students an evident focal question aligned with a problem that would be recognizable to members of the discipline and then engaged them in pursuing that problem. Examples of efforts to engage students in disciplinary inquiry and communication included providing students with opportunities to

- use data collected by climatologists to determine “how and why to categorize tornadoes based on how strong they are?” (grade 6, natural sciences, Video Q1),
- observe a vortex produced with water and two plastic bottles to investigate and construct a scientific explanation of what causes vortices to form (grade 6, integrated science, Video P1),
- read multiple texts about the gold rush and consider, “Was expansion to the west a good or a bad thing?” (grade 8, social studies, Video G1),
- read a chapter of *The Great Gatsby* and consider the relationship between “Nick as the narrator and Fitzgerald as the author” (grade 11, English, Video C1), and
- examine the gross domestic product, literacy rates, and employment rates of various countries to determine, “What might be the relationship [among] these variables?” (grade 12, statistics, Video J1).

In these 12 videos, interns gave students extended opportunities to work with texts to pursue focal questions; typically, this involved working with data or otherwise investigating the focal question using one or more texts and communicating claims in response to the focal question. Texts were both teacher supplied and student

Table 3
Disciplinary literacy teaching coding scheme.

Code	Definition
<i>Engaging in the shared practices of the disciplinary community by:</i>	
Framing a disciplinary problem	Describing the overarching problem space that students will work on or problematizing a topic or concept
Posing a pursuable question	Stating specific focal question that will guide inquiry or enable construction of disciplinary knowledge
Returning to current questions and disciplinary concepts under study	After some exploration, returning to questions to determine progress and refine the inquiry
Using texts to generate questions	Conducting literature reviews or other searches to generate questions or sharpen concepts used to guide inquiry
Working with texts to pursue disciplinary questions	Pursuing disciplinary questions by reading and writing texts (e.g., taking bench notes, analyzing data, synthesizing findings)
Using texts to communicate an argument	Making claims, selecting evidence, or sharing reasoning
Using texts to evaluate claims and reasoning of self and others	Questioning a claim or revising a claim based on evidence or logic
Reframing disciplinary problem using new knowledge	Integrating new understandings into refined or evolved problem statement
<i>Engineering students' access to and use of texts and eliciting students' ideas by:</i>	
Activating/building students' knowledge for reading/writing	Activating students' prior knowledge or building new knowledge for immediate use (e.g., facilitating a warm up activity, lecturing, introducing key vocabulary, facilitating an activity to connect students' experiences or knowledge to concepts under study)
Demonstrating literacy practices and processes	Explicitly modeling reading or writing strategies or routines for engaging with texts to provide an approach for meaning making with texts
Supporting students' use of literacy practices and processes	Supporting students' ongoing meaning making by reminding students of a specific literacy practice they should use when reading, providing students with graphic organizer or set of thinking steps for making meaning with texts, or facilitating metacognitive discussion
Connecting work with texts to a social purpose or community	Naming shared underlying norms, assumptions, purposes, or tools of the community of people who read/produce specific texts
Supporting sustained student conversation using one or more texts	Facilitating extended student discussion, such as providing students with sentence starters for academic discourse to be used throughout a lesson or facilitating an activity that supports students in extended conversation using one or more texts
Highlighting literacy learning goals	Naming literacy learning focus through stating the lesson objective, reminding students of learning goals/purposes, or synthesizing learning accomplishments at end of lesson
Assessing and responding to student understanding	Administering exit slip, conferring with student about a piece of writing, or otherwise assessing and responding to students' ideas explicitly
<i>Examining and analyzing words, vocabulary, language, and representations in text(s) by:</i>	
Examining implicit messages in author's ways with words	Considering author's diction, discursive moves, or rhetorical strategies
Examining functions of language	Considering how a word or representation signals meaning
Examining domain-specific language features	Considering language features, conventions, genres, text types, or representations of a domain (e.g., considering the features of a graph displaying a data pattern)
<i>Evaluating how, when, and why to use particular ways with words by:</i>	
Evaluating utility of discourses, rhetorical strategies, or genres	Considering when specific discourses, rhetorical strategies, or genres are useful or effective for accomplishing a specified purpose and when they are not
Evaluating similarities and differences in ways with words across domains	Considering how texts are constructed and produced across domains or how domains differ in their goals and investigations
Evaluating relationships between texts and contexts or culture	Considering how texts are situated in a particular context or culture (e.g., considering why natural scientists often display empirical data using tables, graphs, and charts)

generated, and they ranged among data tables and charts, novels, primary source documents including images, mathematical symbols, and written arguments.

3.1.2. Extent of engineering students' access to and use of texts

Of the four categories, *engineering* was the most frequently coded and described the most total minutes of video data. In all 23 videos, interns offered instruction clearly designed to elicit and support students' meaning making with texts. Examples of literacy support included interns guiding students to

- freewrite about a topic to activate prior knowledge,
- use a graphic organizer to organize and remember information,
- use a specific literacy practice or process, and,
- discuss and build understanding of topics and texts under study.

The quality and intensity of interns' engineering for disciplinary learning varied by lesson.

3.1.3. Extent of examining and analyzing vocabulary, words, and language of the discipline

Nine videos contained efforts to support students' *examination* of language, linguistic choices, or genre features and conventions. In one English lesson, interns asked students to read sample conclusion paragraphs and identify features of a strong conclusion in an essay, ultimately generating a list that included a "golden line" that summarizes the thesis in a memorable way and presents a "call to action" (Video A1).

In a second example, interns supported high school students to consider potential meanings of three illustrations in a young adult novel (Video M1). Students worked together and individually to notice aspects of the images and analyze relationships among the images and the surrounding prose, and then interns facilitated a short discussion about students' observations.

3.1.4. Extent of evaluating how, why, and when to use particular ways with words

Three videos contained efforts to support students' *consideration* or *evaluation* of ways with words and other forms of representation unique to the discipline under study. In one instance, a

Table 4

Coded video enactments.

Video	Grade	Content Area	Engaging	Engineering	Examining	Evaluating
D1	11	ELA		Yes		
E1	8	Social studies		Yes		
L1	9–11	ELA		Yes		
R1	9	Math		Yes		
R2	9	Math		Yes		
S1	9	Math		Yes		
S2	9	Math		Yes		
A1	8	ELA		Yes	Yes	
B1	8	ELA		Yes	Yes	
G1	8	Social studies		Yes	Yes	
M1	9–11	ELA		Yes	Yes	
C1	11	ELA	Yes	Yes		
C2	11	ELA	Yes	Yes		
F1	8	Social studies	Yes	Yes		
H1	10	Natural sciences	Yes	Yes		
I1	10	Natural sciences	Yes	Yes		
J1	12	Math	Yes	Yes		
P1	6	Natural sciences	Yes	Yes		
Q1	6	Natural sciences	Yes	Yes	Yes	
R3	9	Math	Yes	Yes	Yes	
K1	12	Math	Yes	Yes	Yes	Yes
N1	9	Social studies	Yes	Yes	Yes	Yes
O1	9	Social studies	Yes	Yes	Yes	Yes

Note. Yes = present in the data collected.

Groups C, R, and S co-planned and then subdivided in their enactments.

Number of interns per video ranged from 1 to 4.

social studies intern asked students to read two different sources from the 1960s reflecting the US debate over whether to exit the Vietnam War (Video O1). Students were to consider which argument was more persuasive, including considering the choices each author made in representing their stance. The interns also facilitated a discussion in which groups of students considered how different types of texts can be persuasive based on historical context and message. We coded this as an example of evaluating the utility of discourses and rhetorical strategies.

In a second example, math interns in a statistics classroom led an interactive lecture on similarities between selecting evidence for statistical arguments and literary arguments (e.g., attending to the whole “data set,” considering relevance of evidence for the specific claim) (Video K1). In this moment, the interns pointed out to students that argumentation operates in similar ways across domains. We coded this as an early attempt at designing learning opportunities for students that would help them evaluate and critique ways with words across domains because the interns seemed to be connecting for students their more familiar ways of argumentation with literature to their less familiar ways of argumentation with quantitative data.

3.2. Degree of integration of four dimensions

Our framework leads us not to simply count the number of videos in which each dimension of disciplinary literacy was attempted, but to consider the extent to which episodes reveal integration of those dimensions. Here, we found three evident patterns in the combinations of different literacy teaching moves. We found 11 lessons that included engineering absent engaging, 7 lessons that included engaging and engineering, and 5 lessons that included engaging, engineering, and some amount of examining and evaluating.

3.2.1. Engineering only: literacy instruction absent disciplinary purpose

Eleven lessons included scaffolding students’ comprehension of texts and eliciting displays of content knowledge absent

disciplinary purpose or inquiry. We describe one lesson about historical empathy in an eighth grade social studies classroom as a representative example (Video E1).

In this lesson, one intern attempted to model a strategy for reading a historical letter “with the goal of empathy.” She read aloud the letter to students, paraphrasing after each line and adding annotations on a projected display of the text. Then, interns organized students into groups and gave each group a different letter representing various perspectives of the time (i.e., perspectives of women, soldiers, African Americans). Students worked together to make posters representing the perspective of the author of their letter. At the end of class, interns assigned a writing assignment: “... write your own letter from the perspective of someone during the Civil War. You can choose any perspective you want. You should write from the historical character you choose. It has to be addressed to someone and signed by you.” This writing assignment was based on the RAFT content literacy strategy (Santa, 1988), which prompts students to write from someone else’s perspective, considering what that person would say about a particular topic to a particular audience.

In this lesson, the interns attempted to support students’ literacy learning by (a) modeling the process of annotating and paraphrasing when reading, (b) designing a group task that would potentially encourage students to construct meaning from an unfamiliar primary source, (c) designing an activity in which students would talk to one another about their ideas, and (d) assigning an individual writing task somewhat aligned with their instructional goal.

Yet, within the 80-min class period, 0 min were spent by the interns or students engaging in text-based work in the service of a disciplinary question or problem. In the absence of a real question of history, students did not have a clear reason for annotating and talking about letters from the 1800s. Further, although they may have learned a term for an orientation—historical empathy—that might matter in college history classes or for the work of historians, students were not supported in learning how or why such an orientation would be helpful for constructing knowledge of history with texts alongside others.

3.2.2. *Partial integration: literacy instruction in the service of disciplinary inquiry*

Seven lessons, across disciplines, included engaging students in disciplinary practice and engineering their access to disciplinary texts or tasks. In these enactments, the nature of the “engineering” was notably different from that described in the section above. Rather than supporting students toward comprehending a single text or completing a specific task such as filling in a worksheet, interns who both “engaged” and “engineered” tended to offer literacy support that was framed by the disciplinary inquiry at hand. In what follows, we describe a representative lesson taught by two interns in a tenth grade integrated science class (Video I1).

For this lesson, the interns were charged by their mentor teacher with supporting students to investigate the relationship between intensity of light and distance, a relationship that would eventually allow students to consider the habitability of various planets (i.e., the larger unit-level inquiry frame). The interns began this lesson with two brief activities attempting to investigate the phenomenon of light and distance with students. One activity was to look at images of stars and consider their relative brightness and size (e.g., the sun, Betelgeuse); the second activity was noticing differences in the brightness of light bulbs at different wattages and at different distances in the room. Following the light bulb activity, the interns facilitated a discussion about which light bulbs seemed brighter. They introduced vocabulary terms “luminosity” and “apparent brightness” as students brought up those concepts and explicitly positioned students as scientists when they used specialized language (e.g., “Look, we are astronomers now”).

Following these introductory activities, one intern facilitated students’ lab-based investigation into the relationship between intensity and distance. Students collected data from light bulbs using a computerized measurement program. The goal of this investigation was to engage in multiple trials to collect data and ultimately “record data and generate the simplest formula” to represent the relationship of intensity and distance. Interns provided students a description of a strong lab investigation and write-up, and they also gave a quick orientation to the materials to be used in the lab. Students spent the next hour investigating and collecting data and constructing the formula they believed best represented this relationship, and the interns circulated and assisted groups of students.

In this integrated science lesson, the interns enacted a lesson that allowed for students’ prolonged engagement and inquiry into a scientific phenomenon using multiple texts. By providing opportunities for some open-ended investigation and data collection, interns gave students choice and freedom for scientific inquiry, while they also anticipated and offered various forms of literacy support. The interns supported students’ disciplinary uses of texts by (a) introducing specialized vocabulary and necessary knowledge, (b) creating materials that would support students’ organization and analysis of data, and (c) conferring with small groups about their hypotheses and initial findings to support their lab report writing. Together, these supports and scaffolds assisted students’ investigation and use of specialized language and texts. Because the interns’ goals were focused on scientific inquiry and a scientific process, and because they located their literacy teaching within the scientific frame of the lesson, they avoided pitfalls of presenting reading and writing as generic sets of skills or as activities disconnected from the work of science. Notably, they also did not assume that students would not need any support at all in their work with texts, and they instead anticipated the ways that their adolescent students could benefit from literacy instruction based on the potential literacy demands of their lesson.

3.2.3. *Fuller integration: literacy instruction in the service of disciplinary inquiry that also supports students’ examination and evaluation of language, representations, and cultural practice*

We identified five videos that documented interns working with students to examine and/or evaluate words in the service of disciplinary engagement. These videos were still novice in quality, as they were marked by only brief and sometimes misaligned instructional moments about words and ways with words and representations. However, they were qualitatively different as a set than the other videos in our data set. In what follows, we describe a ninth grade social studies lesson that contained attempts at engaging, engineering, examining, and evaluating words and ways with words (Video N1).

Interns opened the class by reminding students of the upcoming debate—a culminating unit assessment—to be held in a few days’ time: “Should the US Senate have rejected the League of Nations in 1919?” Perhaps to activate students’ background knowledge, one intern asked the students to freewrite on a related question: “from an American perspective, what are the pros and cons of national self-determination?”

After 4 min of student talk, the interns explained, “now that we have some context back in our heads ... today we are going to do an historical investigation about our debate question.” The interns further explained,

something historians do is when they are faced with a question, they find sources and look in the sources to find information to help you come to a conclusion about a question. We are going to be looking at seven sources total. Each group has a source and you will have a graphic organizer—you are going to become an expert about your source. You will answer the questions on the graphic organizer and share out information from the various sources. We will use this ... in your debate on Tuesday.

To model how to use and analyze the sources, one intern thought aloud as she began to read the first text of the set. She said, “the first thing a historian and most scholars do is they find out who wrote it and what is it from, they source it. The first document we are working with is called *The Deluge* ...” She then noted the author’s credentials found in the sourcing line and noted that the source was likely to be reliable. She also described how the source was actually dated just last year, and she said, “so this is not a primary source, but [the author] is using a primary source to understand this time period. And these are all good questions for me to think about: Where is the source coming from, who is making the source, and how can I use the source in this debate?”

Interns then took turns reading the source aloud as students followed along and made notes. They stopped three times to prompt students to briefly summarize while reading, “because the source is so dense and that is why we are reading it together.” During reading, one intern commented on the language used in one line, saying “... [the author] must feel really strongly about this [issue].” At minute 31, the interns returned to the guiding question, prompting students to work together to analyze the source further and to select specific evidence from the text that they could use to argue in the affirmative for the League of Nations.

For most of the remaining time, students worked in small groups to analyze additional historical documents, considering how they related to the questions about the League of Nations and to the other sources they had read, and then they were supported to share their analyses with the whole class. During their small group work, students used a graphic organizer to organize their talk and notes. During the whole class discussion, students added the contributions of other groups to their graphic organizer. In the final 5 min, interns encouraged students to consider logos, ethos, and

pathos as they developed their arguments and to ward against relying on a “slippery slope fallacy” and arguing about alternative histories.

This history lesson revealed attempts at integrating the four dimensions of disciplinary literacy teaching. Throughout, the interns engaged students with historical problems, historical texts, and historical literacy and reasoning practices. The start of the lesson offered the disciplinary frame and historical problem, and the interns introduced a variety of text types. These conditions necessitated students’ use of historical literacy practices to understand the texts and construct assertions. In several instances, the interns made clear moves to support students’ access to the source material. The interns modeled key historical thinking practices and provided graphic organizers to support the reading and analysis of the texts. Several times, the interns named practices of historians as a way to make the disciplinary work visible to students. Like the teaching in the second vignette we shared, interns offered literacy support framed by the disciplinary inquiry at hand.

Yet, we also noted examining and evaluating moves from this group. Interns facilitated students’ examination of language by modeling how to attend to language in sources to better understand tone, argument, and rhetorical techniques. The reading, writing, and speaking tasks that students completed throughout the lesson offered them extended opportunities to practice examining the language of a focal text in relation to the guiding question. Finally, during the closing presentation of logos, pathos, and ethos (features of a “great debate”), interns seemed to be attempting to support students’ knowledge of how to construct their own arguments of history. This portion of the lesson lacked any active engagement by the students and only named the features without offering students an immediate opportunity to make use of these ways with words in their own work and argumentation. However superficial and ephemeral, this moment of instruction still pointed to interns’ attempt to support students’ broader understanding of shared and culturally bound ways with words.

4. Discussion and conclusions

Our study of the earliest teaching enactments of 60 interns enrolled in one teacher education program has yielded insight into the nature of participants’ early attempts to provide literacy instruction and has also informed our continued programmatic reform locally.

First, we observed that all of the enactments revealed important approaches related to disciplinary literacy teaching. We noticed interns using complex texts with students and designing literacy scaffolds of varying types meant to support students’ advanced meaning making with those texts. This was the case whether interns were working in urban, suburban, exurban, or rural classrooms, whether they were working in ELA or social studies or natural science or mathematics classrooms, and whether they were working in middle or high school classrooms.

When we considered the extent to which enactments involved disciplinary literacy teaching, we noticed potentially telling disconnects in interns’ practice. Eleven videos included support for students’ comprehension and content learning, sometimes with attention to language. But, the instruction in these videos was detached from disciplinary purposes or the people who do disciplinary work. Although there is much to be praised in the enactments of the interns whose lessons did not include disciplinary inquiry, we argue that literacy instruction that does not involve disciplinary engagement ultimately falls short of providing students with opportunities for disciplinary literacy learning. The first vignette we offered, a lesson on historical empathy, showed an example of the limits of literacy instruction absent an actual

disciplinary context or problem, although it had many of the trappings of a history literacy lesson. For historians, historical empathy is not an end in and of itself. It is a means to constructing knowledge of history with texts in community with others (Seixas, 1993). By removing the literacy practice from its disciplinary purpose, it is reduced to a mere exercise.

Seven of the remaining videos included moments of engaging and engineering. In these enactments, interns supported students’ access to subject area texts in the service of pursuing recognizably disciplinary purposes. In these seven videos, preservice teachers lifted text-based activity above generic comprehension or content mastery goals to pursue a historically specific investigation.

Five videos included engaging, engineering, and examining and/or evaluating. These videos included opportunities for students to begin to examine and evaluate language and representation in addition to literacy support in the service of disciplinary purposes. These enactments, we theorize, are the closest to the type of teaching that could, over time, support young people to develop critical disciplinary practice. In the third vignette, we documented interns posing a pursuable question of history, offering students an extended opportunity to consider primary sources and generate historical claims, and also offering substantial support for students’ literacy learning through a graphic organizer and explicit modeling. We also documented interns’ attempts to help students examine and evaluate words and ways with words. Thus, the nature of the literacy learning opportunities provided to students in these lessons was substantially different than that of the other lessons we analyzed. Put differently, these five lessons revealed closer integration of the teaching of cognitive literacy processes, disciplinary epistemological processes, linguistic processes, and linguistic and discursive navigation across cultural boundaries.

One major conclusion of this work is that at least some novices—even in their first semester of a preservice teacher preparation program and in their first formal lessons—are capable of designing and enacting instruction that approaches an ambitious conceptualization of disciplinary literacy teaching in their subject areas. This is a meaningful observation, and it suggests that disciplinary literacy teaching is not necessarily out of reach for beginners if they have the appropriate support, a conclusion also generated by Reisman et al. (2019) of teacher candidates’ historical text-based discussions. The types of learning opportunities and support offered to our intern participants in their teacher education program included: a field-based foundations of literacy course that was cohorted by content area, carefully selected mentor teachers who tended to be models of disciplinary literacy teaching in their varied school and community contexts, and a group of university-based course and field instructors who were largely unified in their vision of professional growth goals, including disciplinary literacy teaching at increased scope and scale over time. Insights about the importance of supporting preservice teachers’ learning of disciplinary literacy teaching respond to documented patterns elsewhere of preservice teachers’ resistance to literacy instruction (Scott et al., 2018) and general lack of disciplinary literacy teaching in early enactments (Kavanagh & Rainey, 2017).

Of course, as we have pointed out, even the most integrated attempts at disciplinary literacy teaching in our data set were less sophisticated than what we might expect of skilled veterans’ practice and less sophisticated than the practice our team has observed of many of our mentor teachers (Rainey, Maher, Coupland, Franchi, & Moje, 2017). This insight does not diminish the important learning that is marked by these interns’ video-recorded lessons. We would not expect that novices’ teaching in their first semester of a professional preparation program would closely resemble that of veterans. Rather, pointing out how novices’ literacy teaching relates to an aspirational bar illuminates areas for

continued development and further design, coaching, and intervention by teacher educators. In particular, our data patterns point to the need for more careful design and instruction around the “evaluating” dimension, especially considering how teachers can represent the disciplines as social constructions, highlight ways that disciplinary norms and practices themselves may be unjust, and support students to consider how they may choose to participate in those communities and discourses.

Just as we would not expect novices to look like skilled veterans, we would also not necessarily expect novices’ early teaching attempts to be identical to one another. The range of enactments in our data suggests that even within a program that is highly supportive of learning disciplinary literacy teaching, novices may vary in their attempts to design and enact instruction that consistently aligns with these approaches. We suspect that the variations in patterns we documented could be in part explained by a normative learning trajectory that describes literacy teaching development. To do such highly complex and layered work requires not just depth of knowledge (of the discipline, of students, of texts), but also the ability to anticipate potential challenges that students may face, and the ability to respond in the moment to that which cannot be anticipated. The cognitive demand of such enactment is quite high, and it could very well be the case that novices—those who are least automatic, or fluent, in their practice—are simply unable to attend to so much at once. Over time, as various aspects of the work become more fluent, they may be able to move closer and closer towards consistent and high quality disciplinary literacy teaching.

Another explanation for the variation in interns’ practice is that they entered into their professional preparation program with somewhat differing “apprenticeships of observation” (Lortie, 1975) with regard to literacy teaching and differing amounts of disciplinary knowledge. These two factors, amongst myriad others, could account for differences in their early attempts to put disciplinary literacy teaching into practice. For example, if an intern’s own disciplinary knowledge is not developed to the point that they are able to articulate a question or problem that would drive inquiry in their field, even with support, then their enactment of a lesson may superficially treat disciplinary inquiry or it may be absent altogether. Or, it could be that the translation of a disciplinary problem into one that is developmentally appropriate, able to be investigated, and motivating for students is also a skill that is being developed by novice teachers. Ideally, over time a teacher preparation program would support novices’ further development of their own disciplinary understandings for teaching, so novices might show meaningful growth along this dimension over time. But, teacher educators in the first semester of a program would have the least amount of time to provide such professional preparation, and therefore novices’ teaching enactments would be most likely to reflect their differing backgrounds and knowledge bases.

We suspect that, in fact, each of these explanations contributes to the varied enactments of disciplinary literacy teaching that we observed among our interns. These reflections have prompted us to continue to innovate in our teacher education program so that we may better support interns’ disciplinary literacy teaching to develop more quickly and consistently. Building on this analysis, we have created a set of practical tools to support interns’ learning of disciplinary literacy teaching. One such tool is a user-friendly version of our code book, the Disciplinary Literacy Teaching Tagging Scheme. This tool holds promise for representing and decomposing disciplinary literacy teaching practice and thus offers a host of immediate possibilities for teacher educators. In keeping with our theoretical commitments, we underscore that practice is situated and continually reconstituted; as a result, we caution against viewing the Tagging Scheme as a checklist. Rather, it offers a set of possibilities about how to name some of what novices need to

learn to do in regard to disciplinary literacy teaching in classroom contexts. To highlight the range of instructional possibilities that the Tagging Scheme can describe, and to further support ongoing efforts in our program to represent and decompose complex literacy teaching practice, we have also created a video library of veteran and intern disciplinary literacy teaching that spans content areas, grade levels, and school/community contexts. This library includes cornerstone videos that are digitally tagged according to our Tagging Scheme. We expect that these efforts will further support interns’ learning, in large part because of the ways that these tools are supporting instructors and mentor teachers to establish shared language and instructional goals for novice teacher development within and across subject area programs.

For scholars of teacher education and teacher educators, a potentially provocative takeaway from this study is the unsettling and recasting of what is meant by *ambitious literacy teaching and learning* in the first place. We join with others who have sought to articulate features of ambitious teaching, including those who have developed and used subject-area-specific typologies for advancing preservice teachers’ professional learning (Grossman et al., 2019b; Grossman, Kazemi, Kavanagh, Franke & Dutro, 2019). Our coding scheme represents ambitious teaching somewhat differently, in that it foregrounds the relationships among the teaching of subject matter, literacy, and agentic use of language and literacies. By explicitly naming and valuing the blending of these goals in practice, and by creating and using other tools and pedagogies to support preservice teachers’ learning, we anticipate that locally we will see growth in interns’ capacities to move beyond equitable subject and literacy teaching toward teaching for social justice (Moje, 2007). Although there are multiple useful ways of naming and supporting teaching practice within teacher education, we urge others to consider ways of more explicitly focusing on how teachers may be prepared to support students to develop their own critical stances toward disciplinary norms, conventions, and practices even as they learn to engage them in student-centered and intellectually rich work.

By way of limitations, we acknowledge that our study of interns’ emerging disciplinary literacy teaching may not describe the full range of preservice teacher practice. We anticipate that additional work could be done to further describe disciplinary literacy teaching, especially the examining and evaluating dimensions that we detected less frequently. We add that we did not analyze participants’ instructional reasoning, small group or cohort dynamics, or mentor teachers’ practice as a part of this study, and we acknowledge that there may have been additional influences or explanations to explain the patterns we detected.

Yet, in developing preliminary language for the enactment of disciplinary literacy teaching, we have produced a useful lens for analyzing literacy teaching that incorporates large bodies of extant literacy research brought about from multiple theoretical paradigms and including a range of subject-area and disciplinary traditions, and that can holistically describe a range of early enactments by preservice teachers. Thus, our coding scheme and its companion Tagging Scheme may contribute to others who are seeking to design for and study the literacy teaching of novice or early career teachers, especially when subject matter learning and texts are involved.

The field requires additional research to advance the goal of supporting practitioners as they learn to design rich literacy teaching in secondary classrooms. One important line of research would investigate the normative trajectory of learning to “do” disciplinary literacy teaching over time, and how such a trajectory is best supported by professional preparation components such as programmatic structures and tools and teacher educator pedagogies of investigation and enactment. Such a line would require

analyzing novices' practice over time within specific programs, analyzing novices' practice across multiple programs, and analyzing veteran teacher practice. A related set of questions might consider the extent to which the systematic preparation of novice teachers for disciplinary literacy teaching would support those teachers' further development in their first years of independent practice, especially when offered rich opportunities for in-service professional development designed around similar goals (e.g., Schoenbach, Greenleaf, & Murphy, 2016).

Indeed, there is still much to learn about the possibilities and challenges of preparing novices for independent, contextually sensitive disciplinary literacy teaching. Yet, we are encouraged that such aims are feasible with developed theories of professional learning, articulated goals for professional practice, and carefully designed structures, tools, and pedagogies of teacher education.

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